Why `cellCentredReconstruct` states fail on a propagating TNNP front while `gaussPointODE` works

A diagnosis of ignition and conduction failure in `highOrderElectroActivationFoamImplicitPETSc`

Solver diagnosis note

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Abstract

The solver offers two ways of handling the ionic state variables $\mathbf{s} = (s_1, \ldots, s_N)$ in the high-order ionic-current quadrature: `gaussPointODE` (the legacy path, one stiff ODE integration per Gauss point) and `cellCentredReconstruct` (one ODE per cell centre, states then reconstructed at the Gauss points by a high-order LRE of order `p1`/`p2`/`p3`). On the Niederer *et al.* (2012) benchmark with the TNNP cell model, `cellCentredReconstruct` fails to produce a propagating activation as the time step is refined ($\Delta t = 10^{-5}$ s): the wavefront neither ignites (with the nominal stimulus) nor conducts (even with a stronger stimulus), whereas `gaussPointODE` ignites and propagates on the same mesh and Δt. This note shows, with three controlled discrimination runs and a short mathematical argument, that this is *not* a coding bug, nor a time-scheme (Crank–Nicolson) artefact, nor a Δt-scaling error. It is an intrinsic accuracy limitation of evaluating a strongly non-linear ionic kinetics at the *cell-average* membrane potential: by Jensen's inequality the cell-average sodium activation under-estimates the pointwise average, weakening the regenerative I_{Na} current at the sharp upstroke. Raising the reconstruction order (`p2`, `p3`) does not help, because the order only affects how the already-computed states are interpolated, not the potential that drove the ODE.

1 Governing equations and the two state-integration modes

The monodomain problem solved for the transmembrane potential V_m is

$$\chi C_m \frac{\partial V_m}{\partial t} \;=\; \nabla\cdot(\mathbf{D}\,\nabla V_m) \;-\; I_{\mathrm{ion}}(V_m, \mathbf{s}) \;+\; I_{\mathrm{ext}}, \qquad \frac{\partial \mathbf{s}}{\partial t} \;=\; \mathbf{g}(V_m, \mathbf{s}), \tag{1}$$

where $\mathbf{s}$ collects the N gating variables and ion concentrations of the ten Tusscher–Noble–Noble–Panfilov (TNNP) cell model, $\mathbf{g}$ is its stiff right-hand side, and $I_{\mathrm{ion}} = \phi(V_m, \mathbf{s})$ is the (algebraic) total ionic current. The reaction source of (1) is integrated over each cell Ω_c with a high-order quadrature at points $\{x_q\}$ and normalised weights $\{w_q\}$ ($\sum_q w_q = 1$; the LRE weights already carry the sub-cell volume fractions, so the quadrature returns the cell *average* directly),

$$\overline{I_{\mathrm{ion},\,c}} \;=\; \frac{1}{|\Omega_c|}\int_{\Omega_c} I_{\mathrm{ion}}\,\mathrm{d}\Omega \;\approx\; \sum_q w_q\,\phi\big(V_m(x_q),\, \mathbf{s}(x_q)\big). \tag{2}$$

The membrane potential at the Gauss points, $V_m(x_q)$, is obtained in both modes by the same high-order reconstruction of the cell-centred field. The two modes differ *only* in how the state vector $\mathbf{s}(x_q)$ that enters (2) is obtained.

`gaussPointODE` **(legacy).** The stiff ODE is integrated *independently at every Gauss point*, driven by the locally reconstructed potential:

$$\mathbf{s}_q^{n+1} \;=\; \mathrm{ODEsolve}\big(\mathbf{s}_q^{n},\, V_m(x_q);\, \Delta t\big), \qquad \mathbf{s}(x_q) \equiv \mathbf{s}_q^{n+1}. \tag{3}$$

Cost: N_q stiff integrations per cell.

`cellCentredReconstruct` **(new).** The ODE is integrated *once per cell*, driven by the cell-centre potential $\overline{V}_{m,c} \equiv V_m(x_c)$, and the resulting cell-centred states are then reconstructed to the Gauss points by an LRE of order p:

$$\mathbf{s}_c^{n+1} \;=\; \mathrm{ODEsolve}\big(\mathbf{s}_c^n,\, \overline{V}_{m,c};\, \Delta t\big), \qquad \mathbf{s}(x_q) \equiv \widehat{\mathbf{s}}^{(p)}(x_q) = \mathcal{R}^{(p)}\big[\{\mathbf{s}_{c'}^{n+1}\}_{c' \in \mathrm{stencil}}\big](x_q). \tag{4}$$

Cost: one stiff integration per cell (the motivation: I_{Na}-limited TNNP integrations dominate the run time, so 1 vs N_q ODEs per cell is a large saving).

The essential contrast is already visible in (3)–(4): in `gaussPointODE` the non-linear map $V_m \mapsto \mathbf{s}$ is applied to the *pointwise* $V_m(x_q)$; in `cellCentredReconstruct` it is applied to the *cell-average* $\overline{V}_{m,c}$, and only the (already-computed) result is interpolated.

2 Observed symptom

Benchmark: 2-D slab, structured hexahedra, $\Delta x = 0.5\,\mathrm{mm}$ (40×16), TNNP, a 1.5 mm corner stimulus box of intensity $50\,000\,\mathrm{A/m^3}$ for 2 ms, activation threshold 0 mV, Crank–Nicolson, Picard coupling, RKF45 for the states. With `cellCentredReconstruct` at order `p1` ("CCp1"):

- at $\Delta t = 10^{-4}\,\mathrm{s}$ the activation ignites and propagates (diagonal points activate in sequence);
- at $\Delta t = 10^{-5}\,\mathrm{s}$ *nothing* activates — every diagonal sample, including the stimulus site itself, stays at $t_{\mathrm{act}} = 0$.

A finer (more accurate) time step producing *no* activation is the counter-intuitive signature that must be explained.

3 Discrimination experiments

Three controlled runs isolate the cause. All use the same mesh (40×16, $\Delta x = 0.5\,\mathrm{mm}$), $\Delta t = 10^{-5}\,\mathrm{s}$, endtime 10 ms (long enough: the stimulus site fires at ∼1.3 ms when it triggers). "Ignition" means the stimulus site (diagonal index 0) crosses the threshold; "conduction" means the front advances along the diagonal.

Run	Configuration	Ignition	Conduction	Evidence
A	`gaussPointODE` (`states=NO`), CN	**yes**	**yes**	pt. 0 @ 1.20 ms; pts. 0–4 active, front ∼4.5 mm
—	`cellCentredReconstruct` CCp1, CN (original)	no	—	all $t_{\mathrm{act}} = 0$; max $V_m = -27\,\mathrm{mV}$
B	`cellCentredReconstruct` CCp1, **backwardEuler**	no	—	all $t_{\mathrm{act}} = 0$; max $V_m = -27\,\mathrm{mV}$
C	`cellCentredReconstruct` CCp1, CN, stimulus $\times 4$	**yes**	no	pt. 0 @ 0.60 ms; pts. 0–2 only, front stalls ∼2.3 mm

Table 1: Discrimination runs at $\Delta t = 10^{-5}\,\mathrm{s}$. `gaussPointODE` ignites and conducts; `cellCentredReconstruct` does neither with the nominal stimulus, and even a $4\times$ stimulus only ignites without sustaining conduction.

The three runs rule out the "obvious" suspects and localise the cause:

Not the time scheme.
Run B replaces Crank–Nicolson by the fully dissipative backward Euler and still fails identically ($\max V_m = -27\,\mathrm{mV}$). Crank–Nicolson is exonerated.

Not a Δt-scaling bug.
The θ-scheme is Δt-consistent ($\mathsf{A} = \mathsf{M}/\Delta t - \theta\mathsf{L}$, $\mathsf{B} = \mathsf{M}/\Delta t + (1-\theta)\mathsf{L}$, source weighted by $\theta, (1-\theta)$ with no spurious Δt factor), and the stimulus is applied over a fixed physical window $[t_0, t_0 + 2\,\mathrm{ms}]$ independent of Δt. Both Δt inject the same charge.

It *is* the state mode.
Run A is identical to the failing case except that the states are solved at the Gauss points (`gaussPointODE`). It ignites at 1.20 ms and conducts. The only changed ingredient is (3) vs (4).

4 Root cause: averaging a non-linear kinetics

4.1 The sodium activation is convex at the foot of the upstroke

The fast depolarisation is carried by the sodium current $I_{\mathrm{Na}} = g_{\mathrm{Na}}\, m^3 h\, j\, (V_m - E_{\mathrm{Na}})$, whose activation gate relaxes towards the steep sigmoid

$$m_\infty(V_m) = \Big[1 + \exp\big((V_{1/2} - V_m)/k\big)\Big]^{-1}, \qquad V_{1/2} \approx -40\,\mathrm{mV},\ k \approx 6\,\mathrm{mV}. \tag{5}$$

Below its inflection ($V_m < V_{1/2}$) — exactly the sub-threshold "foot" where ignition is decided — m_∞ is *convex*. Over a cell Ω_c that straddles the wavefront, $V_m(x)$ ranges from resting on the trailing side to strongly depolarised on the leading side.

4.2 Jensen's inequality: `cellCentredReconstruct` under-shoots the reactive current

Let $\langle\cdot\rangle = \sum_q w_q(\cdot)$ (with $\sum_q w_q = 1$) denote the cell-quadrature average, so $\overline{V}_{m,c} = \langle V_m\rangle$. `gaussPointODE` evaluates the kinetics pointwise and averages the *outcome*; `cellCentredReconstruct` averages the *input* first. For the convex m_∞,

$$\underbrace{\langle\, m_\infty(V_m)\,\rangle}_{\texttt{gaussPointODE}\ \text{(effective)}} \quad \geq \quad \underbrace{m_\infty\big(\,\langle V_m\rangle\,\big)}_{\texttt{cellCentredReconstruct}} \quad = \; m_\infty(\overline{V}_{m,c}). \tag{6}$$

Because $I_{\mathrm{Na}} \propto m^3$, the gap in (6) is amplified cubically. `cellCentredReconstruct` therefore systematically *under-estimates* the open-sodium fraction, hence the inward (depolarising) current, precisely where V_m varies most within a cell — at the stimulus edge and along the upstroke. In `gaussPointODE` the Gauss points sitting on the depolarised side of the cell each run their own ODE, see a high local $V_m(x_q)$, open their sodium channels, and contribute a strong inward current to (2); those sub-cell "hot spots" are exactly what `cellCentredReconstruct` averages away before the kinetics ever sees them.

4.3 Two failure modes

1. **Ignition failure.** With the nominal stimulus the site depolarises to only $V_m \approx -27\,\mathrm{mV}$ (Runs orig/B) and relaxes back: the weakened I_{Na} never becomes regenerative, so the all-or-none upstroke does not fire. A 4× stimulus (Run C) forces V_m past threshold at the site (0.60 ms), confirming the nominal stimulus was rendered *effectively sub-threshold* by the averaging.
2. **Conduction block.** Even once ignited (Run C), the front stalls after 2–3 cells. Propagation requires each activating cell to supply enough inward charge (*source*) to bring the downstream cells (*sink*) to threshold — the cardiac "safety factor". By (6), `cellCentredReconstruct` lowers the source at the very front where it is needed, driving the safety factor below unity: a numerically induced source–sink mismatch, i.e. conduction block.

5 Why the coarse time step "worked"

At $\Delta t = 10^{-4}$ s the implicit step advances the depolarisation in one large jump: the reaction/stimulus source acts over a big $\Delta t/\mathsf{M}$ increment, producing a transient over-shoot of $\overline{V}_{m,c}$ that can exceed threshold even with the under-estimated I_{Na}. As $\Delta t \to 0$ the computed trajectory converges to the *true* (accurate) one, which under `cellCentredReconstruct`'s weakened current stays sub-threshold — so the deficiency, masked by coarse-step over-shoot, is revealed. Consequently the $\Delta t = 10^{-4}$ s CCp1 activation is an artefact of temporal under-resolution and its activation times should not be trusted; it is worth checking that they agree with `gaussPointODE` at the same Δt.

6 Why higher reconstruction order does not help

Modes `p2` and `p3` change only the operator $\mathcal{R}^{(p)}$ in (4) — how the cell-centred states $\mathbf{s}_c^{n+1}$ are interpolated to the Gauss points. They do *not* change the fact that $\mathbf{s}_c^{n+1}$ was produced by integrating the ODE with the cell-average input $\overline{V}_{m,c}$. The bias in (6) is committed *inside* the ODE solve, before any reconstruction; a higher-order interpolation merely refines an already-biased quantity. Empirically and by construction, CCp2/CCp3 inherit the same ignition and conduction failure as CCp1 on this front.

7 When `cellCentredReconstruct` is valid, and recommendation

(6) is an equality when V_m is (nearly) linear across the cell, i.e. when the sub-cell variation is small compared with the curvature scale of m_∞. This is the condition

$$h_{\text{cell}} \;\ll\; w_{\text{front}}, \tag{7}$$

the cell size being much smaller than the upstroke width. The cardiac upstroke is $\sim$0.5–1 mm wide; at $h = 0.5\,\text{mm}$ a cell spans the whole front and `cellCentredReconstruct` fails, whereas $h \lesssim 0.1\,\text{mm}$ would resolve it — at a cell count that erodes the very cost advantage `cellCentredReconstruct` was meant to provide.

- **Propagation with a stiff cell model (Niederer/TNNP):** use `gaussPointODE` (`stateIntegrationMode gaussPointODE`, or `states="NO"` in the triad sweep). It resolves the sub-cell kinetics and both ignites and conducts.
- **Smooth reaction fields** (e.g. the manufactured MMS case of the FDA solver, where V_m varies little within a cell): `cellCentredReconstruct` is accurate and delivers its intended $N_q \to 1$ ODE speed-up.
- **Possible hybrid (future work):** integrate the ODE at the Gauss points only where $|\nabla V_m|$ is large (the moving front) and at the cell centre with reconstruction elsewhere (resting/plateau). This would recover most of the speed-up while preserving accuracy at the front.

Bottom line. `cellCentredReconstruct` is not broken and its implementation is correct; it simply trades sub-cell resolution of the ionic kinetics for speed. That trade is safe for smooth reactions but invalid for the steep, self-sustaining upstroke of a real cardiac action potential, where averaging the membrane potential before the sodium kinetics suppresses ignition and blocks conduction. For the Niederer benchmark, `gaussPointODE` is the appropriate mode.